



Amanuel Fesseha

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EDUCATION

University of Georgia

Bachelor of Science, Electrical Engineering

Athens, GA

May 2027

Georgia State University

Associates of Science, Electrical Engineering

Atlanta, GA

May 2024

Cumulative GPA: 3.5/4.0

TECHNICAL SKILLS

- **Software:** Altium Designer, KiCad (schematic/PCB), Autodesk (Fusion 360, Inventor, AutoCAD), MatLab
- **Robotics and ML:** IssacSim, ROS2, Gazebo, URDF/Xacro; PyTorch CNNs and transfer learning
- **Languages:** Python, C/C++, SQL; Tools: Git/GitHub, Jupyter: Java, Python, C
- **Embedded:** Design with ESP32, STM32, nRF52; Sensor integration and IO with I2C, SPI, UART

RELEVANT PROJECT EXPERIENCE

CURO Research Fellowship, *Biomedically Adaptive Leaping System*

Summer 2025 - 2026

- Engineered a bio-inspired leaping robot in Fusion 360 with focus on energy efficiency and actuator performance.
- Built custom brushless motor actuators with FOC control, delivering ~**18 Nm** torque with **25%** of the cost.
- Constructed URDF/Xacro for ROS2 Jazzy and Gazebo simulation, resolving mesh and transform issues.
- Optimized gear ratios and control pipelines to improve locomotion adaptability.

Electronic Test-bench, *ECSE 2920 Design Methodology*

Spring 2026

- Designed a Pi Hat integrating Pi interfaces, analog circuitry, waveform generation, and power distribution.
- Routed a 4-layer mixed-signal PCB in Altium Designer, optimizing for manufacturability and system integration.
- Applied design rules and supported PCB fabrication, assembly, and hardware integration.

Sensor-Integrated Gimbal Headlight Tracker, *ELEE 4230 Sensors and Transducers*

Summer 2026

- Designed an adaptive bicycle headlight position tracking, ambient-light sensing, and two-axis servo control.
- Developed the signal-processing pipeline for sensor data, including hardware linearization, analog filtering, ADC conversion, calibration, averaging, and Kalman filtering for stable real-time control.

RELEVANT WORK EXPERIENCE

Create Lab

Athens, GA

Student Engineer

September 2024 - Present

- Engineered custom products with cross-functional teams, translating client briefs into manufacturable designs.
- Managed project timelines, ensuring efficient delivery of high-quality products within specified deadlines.
- Conducted product testing and provided technical feedback to optimize performance and functionality.
- Worked closely with clients to refine project requirements and align final product designs with their expectations.

Projects

Reaction Latency Evaluation – Upgraded testing platform for gerontology research in UGA PSYC.

- Increased test stimulus speed from **5 mph** to **55 mph**, achieving **11×** improvement in test speed and fidelity
- Developed data logging with dedicated control **GUI**, live data preview and automatic CSV storage of data

Pecan Impact Recording Device – pecan-sized wireless system for impact characterization and data logging.

- Designed a **15 × 30 mm custom PCB** using a **WLCSP**-packaged MCU and a 6-axis IMU for motion sensing
- Integrated a **2.4 GHz RF antenna + matching network**, high-rate sampling and onboard data logging.

INVOLVEMENT EXPERIENCE

UGA Robotics Club, *Member* | Athens, GA

August 2024 - Present

BatBot – Batman-themed **1 lb flipper combat robot** built for a materials-constrained BattleBot competition.

- Designed a flipper mechanism within weight limit, balancing actuation force, traction, and structural durability.
- 3D-printed lightweight components and tuned weight distribution and durability.